

Christopher Iván Ortiz Villafañet

.NET Software Engineer · Applied AI (LLMs / RAG) · Mobile (.NET MAUI)

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SUMMARY

Software Engineer specialized in **.NET, Azure and generative AI**, building enterprise systems end to end. At Alvaco I designed a corporate assistant using **RAG and text-to-SQL on Azure OpenAI**, co-led the **.NET MAUI mobile app** that ~400 technicians use to operate 15,000+ machines across Mexico and Latin America, and built **100% of the front-end** of the company's three core systems. I combine backend architecture (ASP.NET Core, EF Core, SQL Server), LLM security and UI/UX design, working with Git flow, automated tests and GitHub Actions CI. Advanced English (C1).

EXPERIENCE

Alvaco Enterprises — Full-Stack Software Developer (AI & Mobile)

Jan 2026 – Present

Operator of 15,000+ amusement and vending machines across Mexico and Latin America. I own the front-end of the three core systems and lead AI integration into operations.

Alvaco — Generative AI corporate assistant

- Designed and built an assistant on **.NET 10, Semantic Kernel and Azure OpenAI (GPT-4o)** combining **multi-source RAG** (4 knowledge bases, embeddings) with read-only **text-to-SQL** over the SQL Server ERP.
- Shipped it to 4 channels: web chat, **WhatsApp Business (Meta Cloud API)**, the Alvaco360 portal and the mobile app ("Otto" tech support with live handoff to an engineer).
- Implemented a **5-layer security model** for LLM-generated SQL: per-department permissions, token quotas, lexical gate, filtered schema and AST validation (ScriptDom), under a least-privilege DB user.
- Cut token usage by up to **71% per query** and repeat-question latency from **~2 s to under 1 s** with semantic caching, model-free routing and question-aware prompt assembly.
- Built a continuous-improvement agent that proposes knowledge-base entries with human approval, plus a cost, usage and feedback dashboard.
- Set up **GitHub Actions CI** (build, ~280 xUnit tests, vulnerable-package scan, secret detection) and a feature → qa → main flow with semantic versioning.

AlvacoMovil Refacciones — Android app for ~400 technicians and 15,000+ machines (project co-lead)

- Co-led design and development of the **.NET MAUI (MVVM)** app and its **ASP.NET Core 9 API on Azure App Service** (18 controllers, 54 endpoints, JWT), replacing a legacy Java app.
- Built **100% of the front-end** (~34 screens) and its design system: cash collection, spare parts, repairs, routes, audits and a status dashboard.
- Innovation:** beyond financial collection, the app captures and sends **each vending machine's operational data** (status, configuration, incidents and history).
- Built a **remote update system from scratch without Google Play**, with version control and blocking of outdated APKs for strict internal security.
- Offline-first** architecture: 32 SQLite tables, incremental catalog sync and a duplicate-safe send queue. Migrated QR scanning to Google ML Kit and integrated the app with Alvaco.

Alvaco360 (SIA 2.0) — Web ERP for Mexico City HQ and operations across Mexico and Latin America (repo founder, front-end lead in a team of 5)

- Bootstrapped the project and built **100% of the front-end** and UX: visual theme, **Microsoft Entra ID (SSO)** login and a permission-driven home screen.
- Delivered HR modules (employees, positions, schedules, time clock), route-based coin counting, Route Management and the **National Equipment Status** dashboard (coverage, payout, revenue).

- Stack: **ASP.NET MVC 5, Entity Framework 6, Azure SQL** (3 databases, 192 entities), Bootstrap 5, DataTables and Chart.js; integrations with Estafeta, DHL, ZKTeco BioTime, Nayax and Verifone.
- Coordinated team branch integration with Conventional Commits. Maintained legacy PHP systems during the migration.

Grupo Difusión Científica — Junior Developer

Feb 2023 – Jan 2024

- Planned and built a desktop application end to end, including its UI design.
- Implemented security measures to protect private content.

Denedig S.A. de C.V. — UI Developer (Internship)

May 2022 – Dec 2022

- Built the UI of an interactive cultural-themed application with a focus on usability.
- Connected a database to display final scores on a web page.

PERSONAL PROJECT

Pets — Pet safety and location system

2025 – 2026

- **.NET MAUI** app with a **Web API on Azure App Service** and **Azure SQL**. Each pet has a **unique QR code**: when scanned, the owner gets the finder's location via **WhatsApp**. Uses **Google Maps API** to find nearby vets.

TECHNICAL SKILLS

- **Generative AI**: Azure OpenAI (GPT-4o), Semantic Kernel, RAG, embeddings, function calling, text-to-SQL, semantic caching, token optimization
- **Backend**: C#, .NET 10/9, ASP.NET Core Web API, ASP.NET MVC 5, EF Core / EF6, LINQ, JWT, OpenID Connect
- **Mobile**: .NET MAUI, MVVM, offline-first SQLite, OTA updates, Google ML Kit
- **Frontend**: HTML5, CSS3, JavaScript, Razor, Bootstrap 5, jQuery, Chart.js, UI/UX and design systems
- **Data & cloud**: SQL Server, Azure SQL, T-SQL, Azure App Service, Microsoft Entra ID
- **DevOps & testing**: Git / GitFlow, GitHub Actions, Docker, xUnit, NSubstitute
- **Other**: PHP, Python, C++, Unity, Android Studio

EDUCATION & LANGUAGES

B.Eng. in Interactive Software Development — Universidad AMERIKE, Mexico City

Jun 2025

Spanish (native) · English (advanced, C1)